

University of Rwanda- CST

Assignment II Module name: MATH I

December 17, 2024

1. Show that the sequence is monotone /or bounded and determine whether it converges:

$$\begin{array}{lll} \text{(a)} \left\{ n - \frac{1}{n} \right\}_{n=1}^{+\infty} & \text{(c)} \left\{ \frac{2^n}{(n+1)!} \right\}_{n=1}^{+\infty} & \text{(e)} \left\{ \frac{4n-1}{5n+2} \right\}_{n=1}^{+\infty} \\ \text{(b)} \left\{ 2 + \frac{1}{n} \right\}_{n=1}^{+\infty} & \text{(d)} \left\{ \cos \frac{\pi}{2n} \right\}_{n=1}^{+\infty} & 3 \end{array}$$

2. Determine whether the given series converges or diverges by using any appropriate test. If it converges find its value:

$$\begin{array}{ll} \text{(a)} \sum_{n=0}^{\infty} \frac{1}{n^2 + 4n + 3} & \text{(d)} \sum_{n=1}^{\infty} \frac{\sin n}{n^3} \\ \text{(b)} \sum_{n=0}^{\infty} \left(\frac{4}{n^2 + 4n + 3} - 9^{-n+2} \cdot 4^{n+1} \right) & \text{(e)} \sum_{n=1}^{\infty} \frac{n^2}{(2n+1)!} \\ \text{(c)} \sum_{n=1}^{\infty} \frac{4n^2 - n^3}{10 + 2n^3} & \end{array}$$

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3. Solve the following system of equation by using the indicated method.

(a)

$$\begin{cases} 2x_1 + x_2 + x_3 = 4 \\ x_1 - x_2 + 2x_3 = 2 \\ 3x_1 - 2x_2 - x_3 = 0 \end{cases} \quad \text{Inverse method.}$$

(b)

$$\begin{cases} 2x_1 + x_2 + x_3 = 4 \\ x_1 - x_2 + 2x_3 = 2 \\ 3x_1 - 2x_2 - x_3 = 0 \end{cases} \quad \text{Cramer's method.}$$

(c)

$$\begin{cases} 10x_1 + x_2 + x_3 = 12, \\ x_1 + 10x_2 - x_3 = 10, \\ x_1 - 2x_2 + 10x_3 = 9, \end{cases} \quad \text{Gauss Elimination method.}$$

(d)

$$\begin{cases} x + 2y + z - w = -2, \\ 2x + 3y - z + 2w = 7, \\ x + y + 3z - 2w = -6, \\ x + y + z + w = 2. \end{cases} \quad \text{Gauss - Jordan method.}$$

4. Determine the centre, radius, and interval of convergence of each of the following power series.

$$(a) \sum_{n=0}^{\infty} \frac{1}{n} \left(\frac{x+2}{2} \right)^n \quad (b) \sum_{n=0}^{\infty} n^3 (2x-3)^n \quad (c) \sum_{n=0}^{\infty} \frac{(4x-1)^n}{n^n}$$

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Date of submission: December 23, 2024.